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58 GiE      F1      0      VALUE={v(ie)*g}
59 *          voltage components
60 EuxS  usxS  0  VALUE={{(2*v(u1,u2)-v(v1,v2)-v(w1,w2))/3}}
61 EuyS  usyS  0  VALUE={{(v(v1,v2)-v(w1,w2))/sqrt(3)}}
62 Eux    usx    0  VALUE={ v(usxS)*cos(v(alpha))+v(usyS)*sin(v(alpha))}
63 Euy    usy    0  VALUE={-v(usxS)*sin(v(alpha))+v(usyS)*cos(v(alpha))}
64 * Starpoint
65 R2    v2  GND1  1u
66 R12   u2  GND1  1u
67 R23   w2  GND1  1u
68 RGND1 GND1 0 1e6
69 *          Zerosystem
70 Eu0   u_Null 0  VALUE={{(v(u1,u2)+v(v1,v2)+v(w1,w2))/3}}
71 Ei0   i_Null 0  VALUE={idt(v(u_Null)-R0*v(i_null))/LO}
72 *          current components
73 EisxS isxS  0  VALUE={v(isx)*cos(v(alpha))-v(isy)*sin(v(alpha))}
74 EisyS isyS  0  VALUE={v(isy)*cos(v(alpha))+v(isx)*sin(v(alpha))}
75 *          magnetic coupling of direct axis with saturation
76 Gpsix  0  isx  VALUE={v(psisx)}
77 Rsx    isx  0  {1/Lsd}
78 Gpsidx isx iDx1 VALUE={v(psiDx)} ;
79 Rsdx   iDx1 isx {1/LDx}
80 GpsiEx iDx1 imx1 VALUE={v(psiE)}
81 RsEx   imx1 iDx1 {1/LE}
82 vphx   imx1 imxa1 0
83 Eimx   imxa1 0  VALUE={ImNx*(a1*i(vphx)/PsiN+(1-a1)*PWRS(i(vphx)/PsiN,m1))}
84 *          magnetic coupling of quadrature axis with saturation
85 Gpsiy  0  isy  VALUE={v(psisy)}
86 Rsy    isy  0  {1/Lsq}
87 Gpsidy isy imy1 VALUE={v(psiDy)}
88 Rsdy   imy1 isy {1/LDy}
89 vphy   imy1 imya1 0
90 Eimy   imya1 0  VALUE={ImNy*(a2*i(vphy)/PsiN+(1-a2)*PWRS(i(vphy)/PsiN,m2))}
91 *          damper currents
92 Eidx   idx 0  VALUE={v(idx1)-v(isx)}
93 Eidy   idy 0  VALUE={v(imy1)-v(isy)}
94 *          exciter current
95 Eie    ie 0  VALUE={v(imx1)-v(idx1)}
96 *          Flux components
97 Epsi_sx psisx  0  VALUE={err0*PsiN+idt(v(usx)-Rs*v(isx)+p*(2*pi)/60*v(n)*v(psisy))}
98 Epsi_sy psisy  0  VALUE={idt(v(usy)-Rs*v(isy)-p*(2*pi)/60*v(n)*v(psisx))}
99 Epsi_Dx psiDx  0  VALUE={err0*PsiN+idt(-RDx*v(idx))}
100 Epsi_Dy psiDy  0  VALUE={idt(-RDy*v(idy))}
101 Epsi_E  psiE  0  VALUE={err0*PsiN*(1+Streuung)+idt(v(F1,0)*g*RE_factor-RE*RE_factor*g*g*v(ie))}
102 *          Load angle
103 Etheta 0 theta VALUE={idt(180/pi*atan2(v(usx),v(usy)))}
104 *          inner torque
105 Emi    Mi  0  VALUE={1.5*p*(i(vphx)*v(isy)-i(vphy)*v(isx))}
106 *          outhertorque
107 EMoment M  0  VALUE={-v(Mi)+v(Rbg)}
108 Vn    n  0  {n1}
109 * friction
110 GRbg  0  Rbg  TABLE {v(n)} (-1,-1) (1,1)
111 RRbg  Rbg  0  {MN*rbg}
112 * Compute angle alpha
113 Ealpha alpha 0  VALUE={alpha_0*pi/180+idt(v(n))*p/60*(2*pi)}
114 .ENDS

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